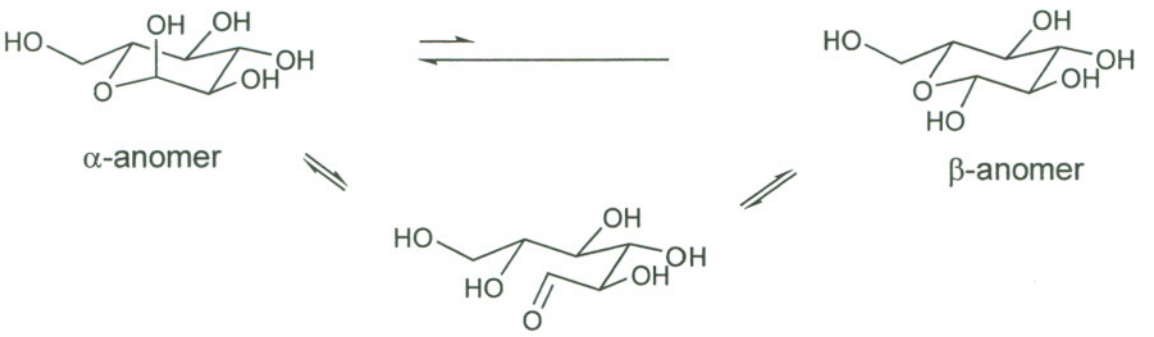
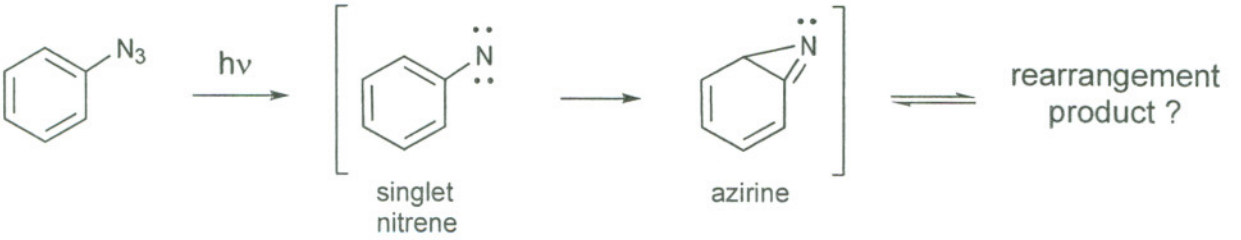
Organic cumulative exam Topics in Physical Organic Chemistry January 19, 2007

1. (40 points) The α- and β-anomers of sugars can interconvert via the open chain hydroxy aldehyde. Using FMO theory, explain why the α-anomer of the sugar below is more stable than the β-anomer.



2. (10 points) Indicate whether the following equilibrium lies toward reactants or products and justify your answer.

3. (10 points) At standard temperature and pressure, what would the difference in free energy have to be for the equilibrium below to favor products by a 100:1 ratio?

4. (40 points) In Meng-Lin Tsao's seminar on Tuesday, he described the laser flash photolysis studies of phenyl azide. One pathway involved the isomerization of the singlet nitrene to a bicyclic azirine. Draw the rearrangement product of the intermediate azirine. What kind of reaction is this (hint: consider the reverse reaction)? Is it a thermally allowed process? Justify your answer using orbital component nomenclature and the generalized orbital symmetry rule.